CLOUD & DEVOPS

GCP Core Services Visual Guide

Compute Engine, Cloud Storage, VPC, IAM, Cloud SQL, BigQuery — how they connect.

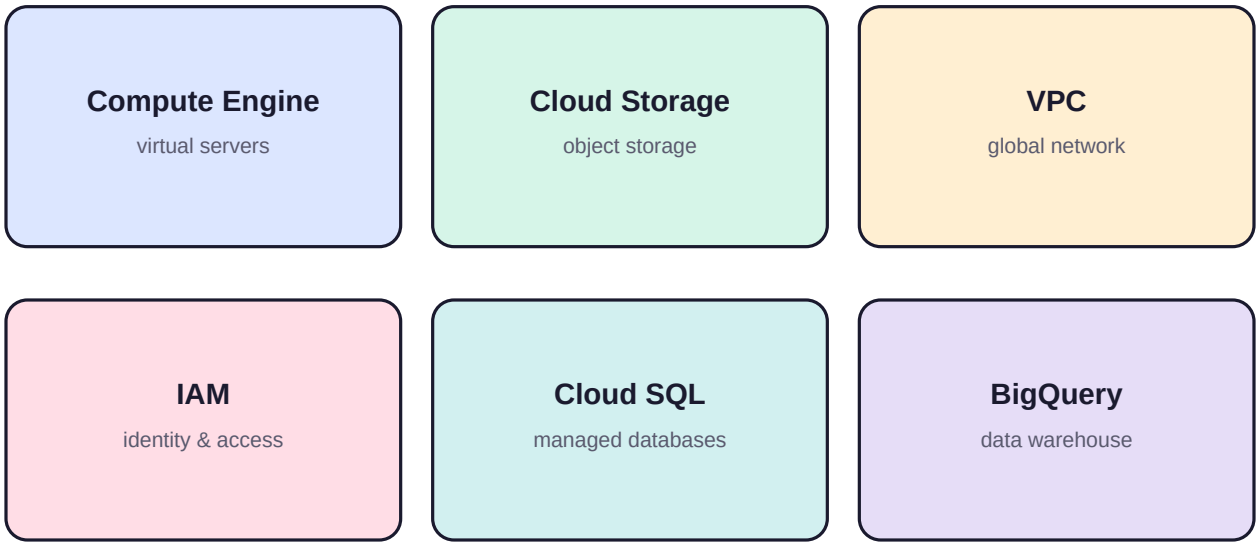
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Meet the Six Core Services

Google's equivalents of the cloud primitives.

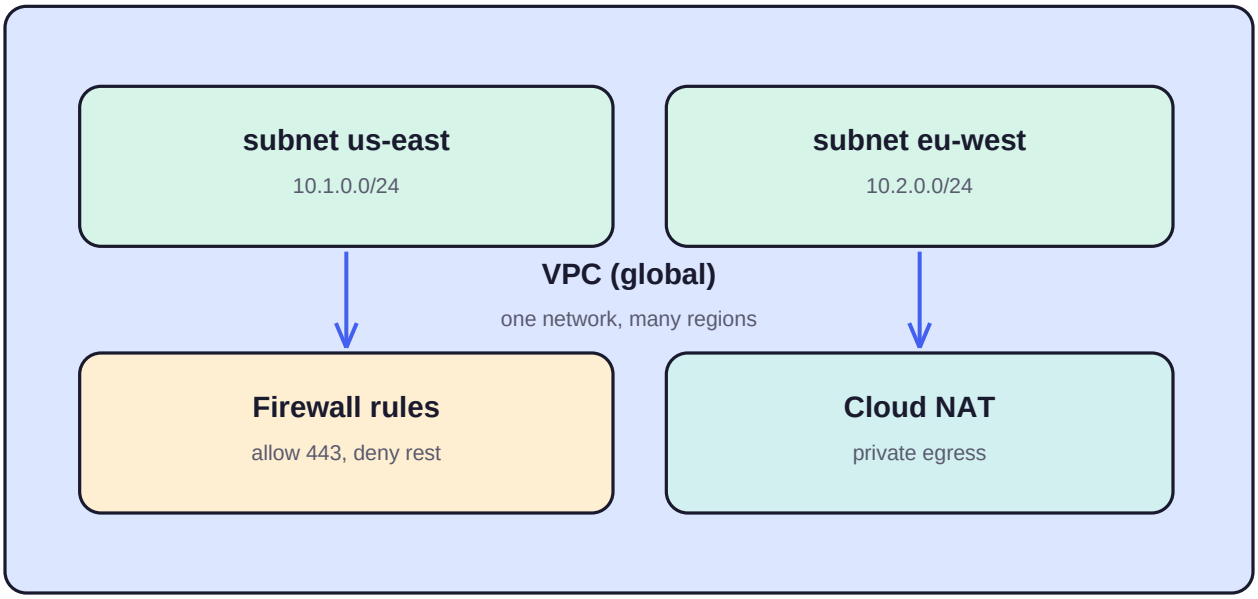


GCP networks are GLOBAL: one VPC spans all regions — subnets are regional, not per-VPC-per-region like AWS.

STEP 2/6

VPC: One Global Network

Single VPC, regional subnets, firewall rules as code.

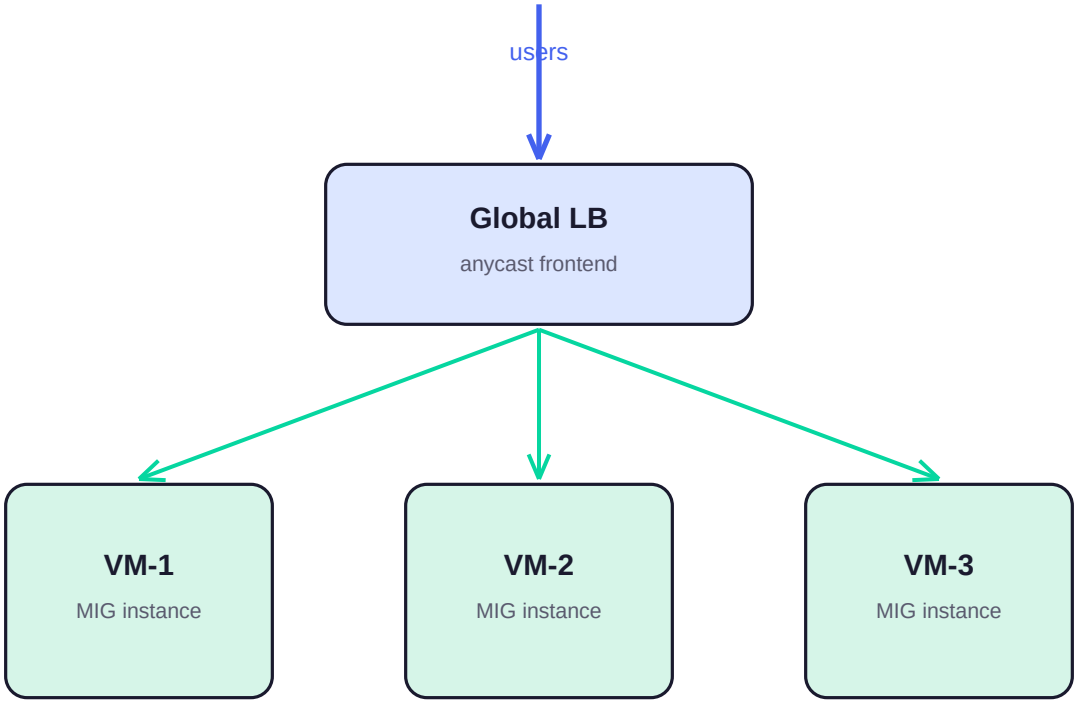


Routes and peering are automatic inside a global VPC — no transit gateways to wire up.

STEP 3/6

The Web Tier: Managed Instance Groups

Compute Engine VMs behind a global load balancer.



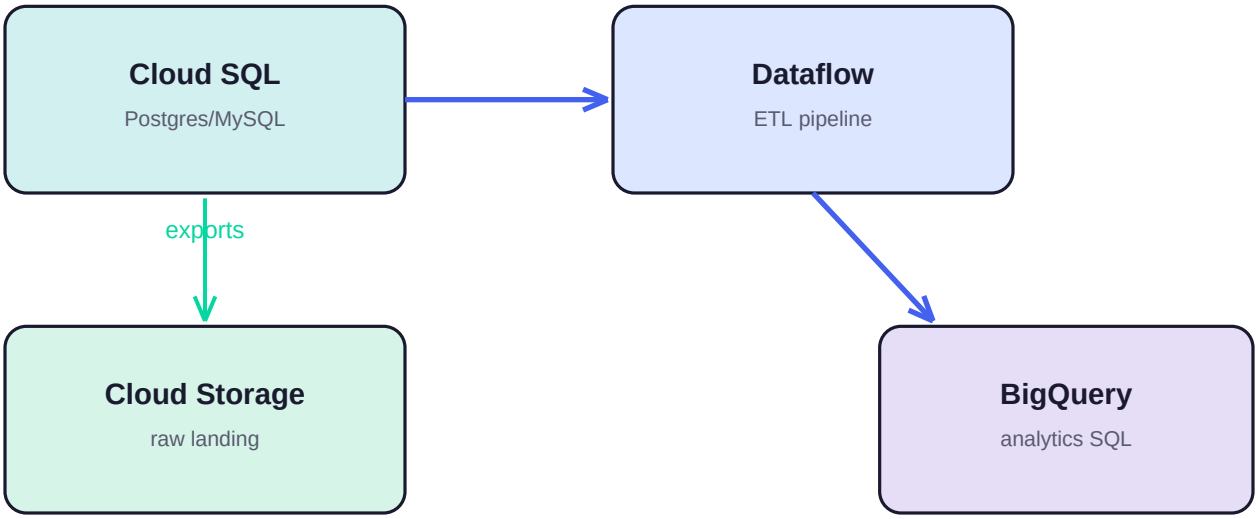
Managed Instance Group (MIG): autoscaling + autohealing, like an ASG.

Global LB terminates TLS at the edge, closest to the user.

STEP 4/6

Data Flow: Cloud SQL → BigQuery

Transactional store feeding the analytics warehouse.

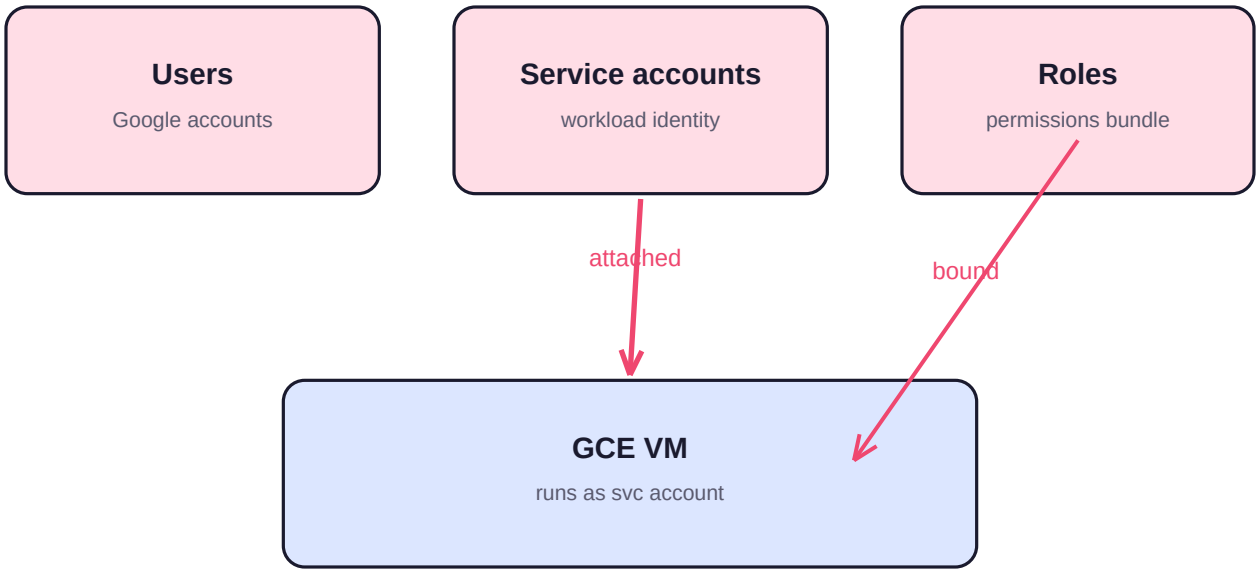


BigQuery is serverless: no clusters to size — you pay per query scanned, not per hour.

STEP 5/6

IAM: Roles & Service Accounts

Workloads get identities too — not just humans.

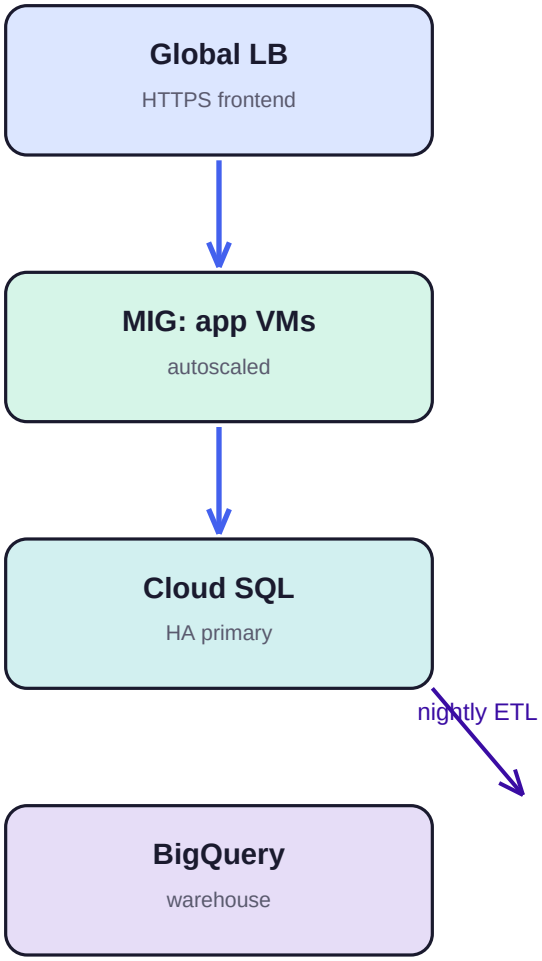


Attach a service account to the VM and grant it a role — code uses default credentials, no keys in the repo.

STEP 6/6

Putting It Together: 3-Tier on GCP

The same architecture, GCP-flavored.



STEP ✓I✓

Key Takeaways

GCP Core Services — Remember

- GCP VPCs are global — subnets are regional, peering is simple by default.
- MIG + Global LB = autoscaled web tier with edge TLS termination.
- Cloud SQL for transactions; BigQuery (serverless) for analytics at scale.
- Service accounts give workloads identities — attach, don't embed keys.
- Firewall rules are stateful and centralized per VPC, not per instance.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev